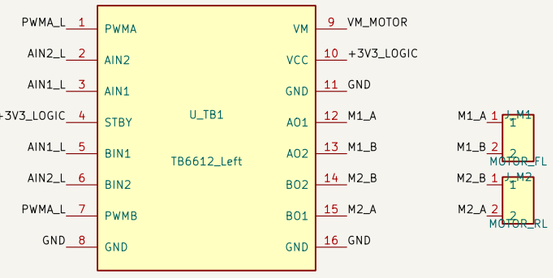
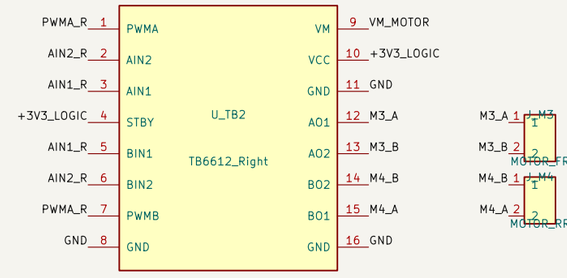
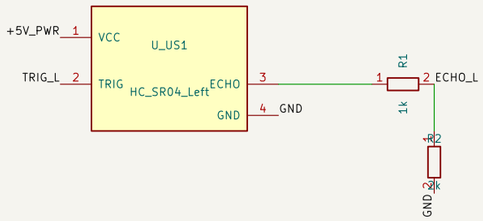
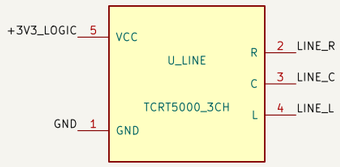
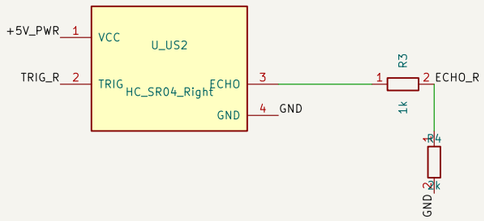

BT1
BAT+
BAT_3S2P_11.1V
GND
SW1
SW_MAIN_10A
FUSE_IN
F1
7.5A_Blade
VM_MOTOR
VM_MOTOR
VM_MOTOR
IN+
IN−
U1
XL4015_Buck_5A
OUT+
OUT−
+5V_PWR
GND
GND
C1
C2
C3
C4
470uF_25V
470uF_25V
470uF_10V
100nF
GND
GND
GND
GND


PWMA_L
AIN2_L
AIN1_L
+3V3_LOGIC
AIN1_L
AIN2_L
PWMA_L
GND
PWMA
AIN2
AIN1
STBY
BIN1
BIN2
PWMB
GND
U_TB1
TB6612_Left
VM
VCC
GND
AO1
AO2
BO2
BO1
GND
VM_MOTOR
+3V3_LOGIC
GND
M1_A
M1_B
M2_B
M2_A
GND
M1
M2
MOTOR_FL
MOTOR_RL

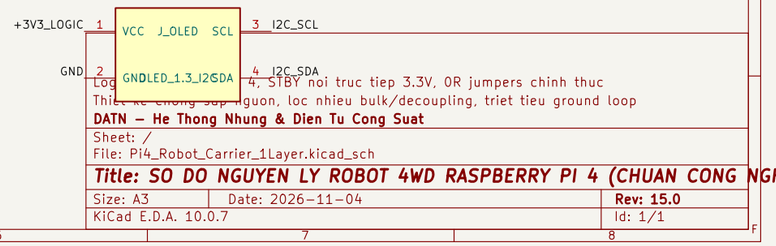

PWMA_R
AIN2_R
AIN1_R
+3V3_LOGIC
AIN1_R
AIN2_R
PWMA_R
GND
PWMA
AIN2
AIN1
STBY
BIN1
BIN2
PWMB
GND
U_TB2
TB6612_Right
VM
VCC
GND
AO1
AO2
BO2
BO1
GND
VM_MOTOR
+3V3_LOGIC
GND
M3_A
M3_B
M4_B
M4_A
GND
M3
M4
MOTOR_FR
MOTOR_RR


+5V_PWR
TRIG_L
VCC
TRIG
HC_SR04_Left
ECHO
GND
U_US1
GND
R1
1k
R2
1k
ECHO_L
GND


+3V3_LOGIC
GND
VCC
U_LINE
TCRT5000_3CH
GND
R
C
L
LINE_R
LINE_C
LINE_L


+5V_PWR
TRIG_R
VCC
TRIG
HC_SR04_Right
ECHO
GND
U_US2
GND
R3
1k
R4
1k
ECHO_R
GND

+3V3_LOGIC
I2C_SDA
I2C_SCL
GND
TRIG_L
ECHO_L
LINE_L
LINE_C
LINE_R
GND
+3V3_PWR
GPIO2_SDA
GPIO3_SCL
GPIO4_AIN1_L
GND
GPIO17_STBY
GPIO27
GPIO22_BIN2_L
+3V3_PWR
GPIO10_LINE_L
GPIO9_LINE_C
GPIO11_LINE_R
GND
U_PI4
Raspberry_Pi_4_40Pin
+5V_PWR
+5V_PWR
GND
GPIO14_TXD
GPIO15_RXD
GPIO18_PWMB_L
GND
GPIO23_TRIG_L
GPIO24_ECHO_L
GND
GPIO25_TRIG_R
GPIO8_CE0
GPIO7_CE1
+5V_PWR
+5V_PWR
GND
TRIG_R
ECHO_R
GND
JMP4
JMP3
JMP2
JMP1
OR
OR
+5V_PWR
+3V3_LOGIC
GND


+3V3_LOGIC
GND
VCC
J_OLED
GNDLED_1.3_I2C
SCL
SDA
I2C_SCL
I2C_SDA
4, STBY noi truc tiep 3.3V, OR jumpers chinh thuc
DATN – He Thong Nhung & Dien Tu Cong Suat
Sheet: /
File: Pi4_Robot_Carrier_1Layer.kicad_sch
Title: SO DO NGUYEN LY ROBOT 4WD RASPBERRY PI 4 (CHUAN CONG NGHIEP
Size: A3
Date: 2026-11-04
Rev: 15.0
KiCad E.D.A. 10.0.7
Id: 1/1